

Joins & Set Operations

2520030030

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1. Creation of database

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
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INSTANCE

- Startup / Shutdown

Query 1 physician SQL File 8" x

Limit to 1000 rows

```
1 • CREATE DATABASE IF NOT EXISTS joins;
2 • USE joins;
```

Output

Action Output

#	Time	Action	Message
✓ 1	19:56:29	CREATE DATABASE IF NOT EXISTS joins	1 row(s) affected
✓ 2	19:56:32	USE joins	0 row(s) affected

2. Create the tables

MySQL Workbench

Local instance MySQL80 x

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- Performance Schema Setup

Query 1 physician SQL File 8" x

Limit to 1000 rows

```
1 • CREATE DATABASE IF NOT EXISTS joins;
2 • USE joins;
3 • CREATE TABLE class (
4     id INT,
5     name VARCHAR(30)
6 );
7 • CREATE TABLE class_info (
8     id INT,
9     address VARCHAR(30)
10 );
```

Output

Action Output

#	Time	Action	Message
✓ 1	19:56:29	CREATE DATABASE IF NOT EXISTS joins	1 row(s) affected
✓ 2	19:56:32	USE joins	0 row(s) affected
✓ 3	19:57:53	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
✓ 4	19:57:56	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected

3. Insertion of data

```
10 ~);
11 • INSERT INTO class VALUES
12 (1, 'ROHIT SAI'),
13 (2, 'Adhvith'),
14 (4, 'Kiran');
15 • INSERT INTO class_info VALUES
16 (1, 'HYDERABAD'),
17 (2, 'DELHI'),
18 (3, 'CHENNAI');
19
```

Output			
Action Output			
#	Time	Action	Message
✓ 1	19:56:29	CREATE DATABASE IF NOT EXISTS joins	1 row(s) affected
✓ 2	19:56:32	USE joins	0 row(s) affected
✓ 3	19:57:53	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
✓ 4	19:57:56	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected
✓ 5	20:00:28	INSERT INTO class VALUES (1, 'ROHIT SAI'), (2, 'Adhvith'), (4, 'Kiran')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 6	20:00:54	INSERT INTO class_info VALUES (1, 'HYDERABAD'), (2, 'DELHI'), (3, 'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

4. CROSS JOIN

```
19 • SELECT *
20 FROM class
21 CROSS JOIN class_info;
22
23
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	id	name	id	address
▶	4	Kiran	1	HYDERABAD
	2	Adhvith	1	HYDERABAD
	1	ROHIT SAI	1	HYDERABAD
	4	Kiran	2	DELHI
	2	Adhvith	2	DELHI
	1	ROHIT SAI	2	DELHI
	4	Kiran	3	CHENNAI
	2	Adhvith	3	CHENNAI
	1	ROHIT SAI	3	CHENNAI

Result 1 x

Output

Action Output

#	Time	Action	Message
✓ 1	19:56:29	CREATE DATABASE IF NOT EXISTS joins	1 row(s) affected
✓ 2	19:56:32	USE joins	0 row(s) affected
✓ 3	19:57:53	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
✓ 4	19:57:56	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected
✓ 5	20:00:28	INSERT INTO class VALUES (1, 'ROHIT SAI'), (2, 'Adhvith'), (4, 'Kiran')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 6	20:00:54	INSERT INTO class_info VALUES (1, 'HYDERABAD'), (2, 'DELHI'), (3, 'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 7	20:01:35	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	9 row(s) returned

5. INNER JOIN

Limit to 1000 rows

```
22 • SELECT *
23 FROM class
24 INNER JOIN class_info
25 ON class.id = class_info.id;
```

Result Grid

	id	name	id	address
1	ROHIT SAI	1	HYDERABAD	
2	Adhvith	2	DELHI	

Result 2 x

Output

Action Output

#	Time	Action	Message
✓ 1	19:56:29	CREATE DATABASE IF NOT EXISTS joins	1 row(s) affected
✓ 2	19:56:32	USE joins	0 row(s) affected
✓ 3	19:57:53	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
✓ 4	19:57:56	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected
✓ 5	20:00:28	INSERT INTO class VALUES (1, 'ROHIT SAI'), (2, 'Adhvith'), (4, 'Kiran')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 6	20:00:54	INSERT INTO class_info VALUES (1, 'HYDERABAD'), (2, 'DELHI'), (3, 'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 7	20:01:35	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	9 row(s) returned
✓ 8	20:03:11	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	2 row(s) returned

6. INNER JOIN — selected columns

```
25 ON class.id = class_info.id;
26 • SELECT class.name, class_info.address
27 FROM class
28 INNER JOIN class_info
29 ON class.id = class_info.id;
```

Result Grid

	name	address
1	ROHIT SAI	HYDERABAD
2	Adhvith	DELHI

Result 3 x

Output

Action Output

#	Time	Action	Message
✓ 1	19:56:29	CREATE DATABASE IF NOT EXISTS joins	1 row(s) affected
✓ 2	19:56:32	USE joins	0 row(s) affected
✓ 3	19:57:53	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
✓ 4	19:57:56	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected
✓ 5	20:00:28	INSERT INTO class VALUES (1, 'ROHIT SAI'), (2, 'Adhvith'), (4, 'Kiran')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 6	20:00:54	INSERT INTO class_info VALUES (1, 'HYDERABAD'), (2, 'DELHI'), (3, 'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 7	20:01:35	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	9 row(s) returned
✓ 8	20:03:11	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	2 row(s) returned
✓ 9	20:05:02	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	2 row(s) returned

7. NATURAL JOIN

```
30 • SELECT *
31 FROM class
32 NATURAL JOIN class_info;
33
34
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	id	name	address
▶	1	ROHIT SAI	HYDERABAD
	2	Adhvith	DELHI

Result 4 x

Output

Action Output

#	Time	Action	Message
✓	2	19:56:32 USE joins	0 row(s) affected
✓	3	19:57:53 CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
✓	4	19:57:56 CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected
✓	5	20:00:28 INSERT INTO class VALUES (1, 'ROHIT SAI'), (2, 'Adhvith'), (4, 'Kiran')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓	6	20:00:54 INSERT INTO class_info VALUES (1, 'HYDERABAD'), (2, 'DELHI'), (3, 'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓	7	20:01:35 SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	9 row(s) returned
✓	8	20:03:11 SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	2 row(s) returned
✓	9	20:05:02 SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT ...	2 row(s) returned
✓	10	20:06:38 SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	2 row(s) returned

8. Add more data for OUTER JOINS

Limit to 1000 rows

```
33 • INSERT INTO class VALUES
34 (5, 'Nikhil');
35 • INSERT INTO class_info VALUES
36 (7, 'BENGALURU'),
37 (8, 'PUNE');
```

Output

Action Output

#	Time	Action	Message
✓	1	19:56:29 CREATE DATABASE IF NOT EXISTS joins	1 row(s) affected
✓	2	19:56:32 USE joins	0 row(s) affected
✓	3	19:57:53 CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
✓	4	19:57:56 CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected
✓	5	20:00:28 INSERT INTO class VALUES (1, 'ROHIT SAI'), (2, 'Adhvith'), (4, 'Kiran')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓	6	20:00:54 INSERT INTO class_info VALUES (1, 'HYDERABAD'), (2, 'DELHI'), (3, 'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓	7	20:01:35 SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	9 row(s) returned
✓	8	20:03:11 SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	2 row(s) returned
✓	9	20:05:02 SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0,...	2 row(s) returned
✓	10	20:06:38 SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	2 row(s) returned
✓	11	20:09:01 INSERT INTO class VALUES (5, 'Nikhil')	1 row(s) affected
✓	12	20:09:04 INSERT INTO class_info VALUES (7, 'BENGALURU'), (8, 'PUNE')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0

9. LEFT OUTER JOIN

Query 1 physician SQL File 8* x

```
38 • SELECT *
39 FROM class
40 LEFT OUTER JOIN class_info
41 ON class.id = class_info.id;
42
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	id	name	id	address
▶	1	ROHIT SAI	1	HYDERABAD
	2	Adhvith	2	DELHI
	4	Kiran	NULL	NULL
	5	Nikhil	NULL	NULL

10. LEFT JOIN

Query 1 physician SQL File 8* x

```
42 • SELECT *
43 FROM class
44 LEFT JOIN class_info
45 ON class.id = class_info.id
46 WHERE class_info.id IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	id	name	id	address
▶	4	Kiran	NULL	NULL
	5	Nikhil	NULL	NULL

11 — RIGHT OUTER JOIN

MySQL Workbench

Local instance MySQL80 x

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Query 1 physician SQL File 8* x

```
47 • SELECT *
48 FROM class
49 RIGHT OUTER JOIN class_info
50 ON class.id = class_info.id;
51
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	id	name	id	address
▶	1	ROHIT SAI	1	HYDERABAD
	2	Adhvith	2	DELHI
	NULL	NULL	3	CHENNAI
	NULL	NULL	7	BENGALURU
	NULL	NULL	8	PUNE

Result 8 x

12 — UNION

MySQL Workbench

Local instance MySQL80 x

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Administration Schemas

Information

No object selected

Query 1 physician SQL File 8*

Limit to 1000 rows

```
51 CREATE TABLE first_table (  
52     id INT,  
53     name VARCHAR(30)  
54 );  
55 CREATE TABLE second_table (  
56     id INT,  
57     name VARCHAR(30)  
58 );
```

Output

Action Output

#	Time	Action	Message
6	20:00:54	INSERT INTO class_info VALUES (1, 'HYDERABAD'), (2, 'DELHI'), (3, 'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
7	20:01:35	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	9 row(s) returned
8	20:03:11	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	2 row(s) returned
9	20:05:02	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	2 row(s) returned
10	20:06:38	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	2 row(s) returned
11	20:09:01	INSERT INTO class VALUES (5, 'Nikhil')	1 row(s) affected
12	20:09:04	INSERT INTO class_info VALUES (7, 'BENGALURU'), (8, 'PUNE')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
13	20:10:12	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	4 row(s) returned
14	20:12:43	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT 0, 1000	2 row(s) returned
15	20:13:18	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT 0, 1000	2 row(s) returned
16	20:19:10	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
17	20:20:07	CREATE TABLE first_table (id INT, name VARCHAR(30))	0 row(s) affected
18	20:20:11	CREATE TABLE second_table (id INT, name VARCHAR(30))	0 row(s) affected

Query 1 physician SQL File 8*

Limit to 1000 rows

```
59 INSERT INTO first_table VALUES  
60     (1, 'Mohammad'),  
61     (2, 'Rohit Sai');  
62 INSERT INTO second_table VALUES  
63     (2, 'Rohit Sai'),  
64     (3, 'NTR');
```

Output

Action Output

#	Time	Action	Message
1	19:56:29	CREATE DATABASE IF NOT EXISTS joins	1 row(s) affected
2	19:56:32	USE joins	0 row(s) affected
3	19:57:53	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
4	19:57:56	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected
5	20:00:28	INSERT INTO class VALUES (1, 'ROHIT SAI'), (2, 'Adhithi'), (4, 'Kiran')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
6	20:00:54	INSERT INTO class_info VALUES (1, 'HYDERABAD'), (2, 'DELHI'), (3, 'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
7	20:01:35	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	9 row(s) returned
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11	20:09:01	INSERT INTO class VALUES (5, 'Nikhil')	1 row(s) affected
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14	20:12:43	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT 0, 1000	2 row(s) returned
15	20:13:18	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT 0, 1000	2 row(s) returned
16	20:19:10	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
17	20:20:07	CREATE TABLE first_table (id INT, name VARCHAR(30))	0 row(s) affected
18	20:20:11	CREATE TABLE second_table (id INT, name VARCHAR(30))	0 row(s) affected
19	20:21:33	INSERT INTO first_table VALUES (1, 'Mohammad'), (2, 'Rohit Sai')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
20	20:21:35	INSERT INTO second_table VALUES (2, 'Rohit Sai'), (3, 'NTR')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0

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Administration Schemas

Information

No object selected

Object Info Session

Query 1 physician SQL File 8"

Limit to 1000 rows

```

65 SELECT * FROM first_table
66 UNION
67 SELECT * FROM second_table;
68
69

```

Result Grid

id	name
1	Mohammad
2	Rohit Sai
3	NTR

Output

Action Output

#	Time	Action	Message
6	20:00:54	INSERT INTO class_info VALUES (1, 'HYDERABAD'), (2, 'DELHI'), (3, 'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
7	20:01:35	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	9 row(s) returned
8	20:03:11	SELECT * FROM class INNER JOIN class_info ON class_id = class_info.id LIMIT 0, 1000	2 row(s) returned
9	20:05:02	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class_id = class_info.id LIMIT ...	2 row(s) returned
10	20:06:38	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	2 row(s) returned
11	20:09:01	INSERT INTO class VALUES (5, 'Nikhil')	1 row(s) affected
12	20:09:04	INSERT INTO class_info VALUES (7, 'BENGALURU'), (8, 'PUNE')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
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15	20:13:18	SELECT * FROM class LEFT JOIN class_info ON class_id = class_info.id WHERE class_info.id IS NULL LIMIT ...	2 row(s) returned
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17	20:20:07	CREATE TABLE first_table (id INT, name VARCHAR(30))	0 row(s) affected
18	20:20:11	CREATE TABLE second_table (id INT, name VARCHAR(30))	0 row(s) affected
19	20:21:33	INSERT INTO first_table VALUES (1, 'Mohammad'), (2, 'Rohit Sai')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
20	20:21:35	INSERT INTO second_table VALUES (2, 'Rohit Sai'), (3, 'NTR')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
21	20:22:27	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned

13 — UNION ALL

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Information

No object selected

Object Info Session

Query 1 physician SQL File 8"

Limit to 1000 rows

```

68 SELECT * FROM first_table
69 UNION ALL
70 SELECT * FROM second_table;
71
72

```

Result Grid

id	name
1	Mohammad
2	Rohit Sai
2	Rohit Sai
3	NTR

Result 10 x

Output

Action Output

#	Time	Action	Message
7	20:01:35	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	9 row(s) returned
8	20:03:11	SELECT * FROM class INNER JOIN class_info ON class_id = class_info.id LIMIT 0, 1000	2 row(s) returned
9	20:05:02	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class_id = class_info.id LIMIT ...	2 row(s) returned
10	20:06:38	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	2 row(s) returned
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12	20:09:04	INSERT INTO class_info VALUES (7, 'BENGALURU'), (8, 'PUNE')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
13	20:10:12	SELECT * FROM class LEFT OUTER JOIN class_info ON class_id = class_info.id LIMIT 0, 1000	4 row(s) returned
14	20:12:43	SELECT * FROM class LEFT JOIN class_info ON class_id = class_info.id WHERE class_info.id IS NULL LIMIT ...	2 row(s) returned
15	20:13:18	SELECT * FROM class LEFT JOIN class_info ON class_id = class_info.id WHERE class_info.id IS NULL LIMIT ...	2 row(s) returned
16	20:19:10	SELECT * FROM class RIGHT OUTER JOIN class_info ON class_id = class_info.id LIMIT 0, 1000	5 row(s) returned
17	20:20:07	CREATE TABLE first_table (id INT, name VARCHAR(30))	0 row(s) affected
18	20:20:11	CREATE TABLE second_table (id INT, name VARCHAR(30))	0 row(s) affected
19	20:21:33	INSERT INTO first_table VALUES (1, 'Mohammad'), (2, 'Rohit Sai')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
20	20:21:35	INSERT INTO second_table VALUES (2, 'Rohit Sai'), (3, 'NTR')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
21	20:22:27	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned
22	20:23:24	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned

14 — INTERSECT

MySQL Workbench

Local instance MySQL80 x

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Information

No object selected

Object Info Session

Query 1 physician SQL File 8* x

Limit to 1000 rows

```
71 • SELECT * FROM first_table
72 INTERSECT
73 SELECT * FROM second_table;
```

Result Grid

	id	name
▶ 2		Rohit Sai

Result 11 x

Output

Action Output

#	Time	Action	Message
✓ 8	20:03:11	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	2 row(s) returned
✓ 9	20:05:02	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000
✓ 10	20:06:38	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	
✓ 11	20:09:01	INSERT INTO class VALUES (5, 'Nikhil')	
✓ 12	20:09:04	INSERT INTO class_info VALUES (7, 'BENGALURU'), (8, 'PUNE')	records: 2 Duplicates: 0 Warnings: 0
✓ 13	20:10:12	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	4 row(s) returned
✓ 14	20:12:43	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT 0, 1000	2 row(s) returned
✓ 15	20:13:18	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT 0, 1000	2 row(s) returned
✓ 16	20:19:10	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
✓ 17	20:20:07	CREATE TABLE first_table (id INT, name VARCHAR(30))	0 row(s) affected
✓ 18	20:20:11	CREATE TABLE second_table (id INT, name VARCHAR(30))	0 row(s) affected
✓ 19	20:21:33	INSERT INTO first_table VALUES (1, 'Mohammad'), (2, 'Rohit Sai')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓ 20	20:21:35	INSERT INTO second_table VALUES (2, 'Rohit Sai'), (3, 'NTR')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓ 21	20:22:27	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned
✓ 22	20:23:24	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned
✓ 23	20:26:54	SELECT * FROM first_table INTERSECT SELECT * FROM second_table	1 row(s) returned

16 — common names

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

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Administration Schemas

Information

No object selected

Object Info Session

Query 1 physician SQL File 8*

Limit to 1000 rows

```
74 SELECT name FROM first_table
75 INTERSECT
76 SELECT name FROM second_table
```

Result Grid

	name
▶	Rohit Sai

Result 12 x

Output

Action Output

#	Time	Action	Message
✓	9 20:05:02	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	2 row(s) returned
✓	10 20:06:38	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	2 row(s) returned
✓	11 20:09:01	INSERT INTO class VALUES (5, 'Nikhil')	1 row(s) affected
✓	12 20:09:04	INSERT INTO class_info VALUES (7, 'BENGALURU'), (8, 'PUNE')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓	13 20:10:12	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	2 row(s) returned
✓	14 20:12:43	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class.id = class_info.id	2 row(s) returned
✓	15 20:13:18	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class.id = class_info.id	2 row(s) returned
✓	16 20:19:10	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	2 row(s) returned
✓	17 20:20:07	CREATE TABLE first_table (id INT, name VARCHAR(30))	0 row(s) affected
✓	18 20:20:11	CREATE TABLE second_table (id INT, name VARCHAR(30))	0 row(s) affected
✓	19 20:21:33	INSERT INTO first_table VALUES (1, 'Mohammad'), (2, 'Rohit Sai')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓	20 20:21:35	INSERT INTO second_table VALUES (2, 'Rohit Sai'), (3, 'NTR')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓	21 20:22:27	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned
✓	22 20:23:24	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned
✓	23 20:26:54	SELECT * FROM first_table INTERSECT SELECT * FROM second_table	1 row(s) returned
✓	24 20:27:38	SELECT name FROM first_table INTERSECT SELECT name FROM second_table	1 row(s) returned

17 - MINUS

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

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Administration Schemas

Information

No object selected

Query 1 physician SQL File 8*

Limit to 1000 rows

```
77 SELECT first_table.*
78 FROM first_table
79 LEFT JOIN second_table
80 ON first_table.id = second_table.id
81 WHERE second_table.id IS NULL;
82
```

Result Grid

	id	name
▶	1	Mohammad

Result 13 x

Output

Action Output

#	Time	Action	Message
✓	20 20:21:35	INSERT INTO second_table VALUES (2, 'Rohit Sai'), (3, 'NTR')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓	21 20:22:27	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned
✓	22 20:23:24	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned
✓	23 20:26:54	SELECT * FROM first_table INTERSECT SELECT * FROM second_table	1 row(s) returned
✓	24 20:27:38	SELECT name FROM first_table INTERSECT SELECT name FROM second_table	1 row(s) returned
✗	25 20:29:09	SELECT * FROM first_table MINUS SELECT * FROM second_table	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'MINUS SELECT * FROM second_table' at line 1
✗	26 20:30:41	SELECT * FROM first_table MINUS SELECT * FROM second_table	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'MINUS SELECT * FROM second_table' at line 1
✓	27 20:32:08	SELECT first_table.* FROM first_table LEFT JOIN second_table ON first_table.id = second_table.id WHERE se...	1 row(s) returned

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator Query 1 physician SQL File 8* x

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Administration Schemas

Information

No object selected

```
82 • SELECT first_table.name
83 FROM first_table
84 LEFT JOIN second_table
85 ON first_table.name = second_table.name
86 WHERE second_table.name IS NULL;
87 --
```

Result Grid

	name
▶	Mohammad

Result 14 x

Output

Action Output

#	Time	Action	Message
✓	21 20:22:27	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned
✓	22 20:23:24	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned
✓	23 20:26:54	SELECT * FROM first_table INTERSECT SELECT * FROM second_table	1 row(s) returned
✓	24 20:27:38	SELECT name FROM first_table INTERSECT SELECT name FROM second_table	1 row(s) returned
✗	25 20:29:09	SELECT * FROM first_table MINUS SELECT * FROM second_table	Error Code: 1064
✗	26 20:30:41	SELECT * FROM first_table MINUS SELECT * FROM second_table	Error Code: 1064
✓	27 20:32:08	SELECT first_table.* FROM first_table LEFT JOIN second_table ON first_table.id = second_table.id WHERE se...	1 row(s) returned
✓	28 20:33:15	SELECT first_table.name FROM first_table LEFT JOIN second_table ON first_table.name = second_table.name ...	1 row(s) returned

18 - Full Outer Join

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator Query 1 physician SQL File 8* x

Limit to 1000 rows

```
87 • SELECT *
88 FROM class
89 LEFT JOIN class_info
90 ON class.id = class_info.id
91 WHERE class_info.id IS NULL
92 UNION
93 SELECT *
94 FROM class
95 RIGHT JOIN class_info
96 ON class.id = class_info.id
97 WHERE class.id IS NULL;
98
```

Result Grid

	id	name	id	address
▶	4	Kiran	NULL	NULL
	5	Nikhil	NULL	NULL
	NULL	NULL	3	CHENNAI
	NULL	NULL	7	BENGALURU
	NULL	NULL	8	PUNE

Administration Schemas

Information

No object selected

Result 15 x

Output

Action Output

#	Time	Action	Message
✗	25 20:29:09	SELECT * FROM first_table MINUS SELECT * FROM second_table	Error Code: 1064, You
✗	26 20:30:41	SELECT * FROM first_table MINUS SELECT * FROM second_table	Error Code: 1064, You
✓	27 20:32:08	SELECT first_table.* FROM first_table LEFT JOIN second_table ON first_table.id = second_table.id WHERE se...	1 row(s) returned
✓	28 20:33:15	SELECT first_table.name FROM first_table LEFT JOIN second_table ON first_table.name = second_table.name ...	1 row(s) returned
✓	29 20:34:29	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL UNIO...	5 row(s) returned